

Henry Court

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in henry-c2908

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🌐 Portfolio

Education

University of Bristol

Sep 2023 - Jun 2027

MEng in Computer Science

- Maintaining a First-Class average of 76% across my first three years of study
- **Key Modules:** Machine Learning, Computer Graphics, Teams Project, Algorithms and Data

Oldfield School, Bath

Sep 2016 - Jun 2023

- A Levels: A*A*AA in Maths, Further Maths, Computer Science and Physics
- GCSES: Grades 7-9 in Core Subjects, Spanish, Geography, Computer Science and Creative iMedia

Experience

Web Game Development Intern - *OceanX*

Aug 2026 - Present

- Developed a proof-of-concept web game in Unity, adapting an interactive exhibit from Singapore's ArtScience Museum
- Led a five-person multidisciplinary student team as Project Manager, delivering a proof-of-concept within a five week, 250-person-hour development period

Software Development Intern - *University of Bristol*

May 2025 - Jun 2025

- Extended a Springboot-based CRM system in a student team for the University of Bristol's Industrial Liaison Office
- Implemented Spring Security with role-based access levels and authentication, and developed a dynamic user interface to improve usability and user experience

Software Engineering Project Mentor - *University of Bristol*

Sep 2025 - Jun 2026

- Mentored a team of five second-year students throughout their Software Engineering Project, providing weekly advice and feedback; the team achieved a first class mark
- Assessed ten project teams over two formative vivas, evaluating presentations and challenging students on technical choices in preparation for summative assessment

Student Representative - *University of Bristol*

Sep 2023 - Present

- Elected Course Representative for four consecutive years, representing a cohort of ~150 students and driving curriculum changes including a Computer Architecture content overhaul and a second-year curriculum restructure
- Elected School Representative for two consecutive years, contributing Computer Science student perspectives in faculty-level decision-making

Waiter, Bartender and Kitchen Porter - *Queen and Whippet*

Jul 2023 - Present

- Worked across front-of-house, bar and kitchen operations, serving food and drinks at events with 200+ guests

Projects

Monster Hunter Loadout Planner

Java, Spring Boot, Thymeleaf

- Built an interactive web app for planning and optimising armor loadout skills based on JSON data imported from Monster Hunter Rise

Nonogram Solver

Python

- Built a universal application to solve any Nonogram, also known as a Japanese crossword
- Designed a logical deduction algorithm, and a guess → contradiction backtracking algorithm for advanced puzzles

VR Horror Game

Unity, C#

- Led VR development for Project: Asphyxia, coordinating VR implementation within a team of 7 students
- Mitigated motion sickness by researching, implementing, and iterating a novel VR locomotion system

Cornell Box Computer Graphics Engine

C++

- Created a graphics engine to render 3D models in 2D space, such as the cornell box model
- Included features such as soft shadows, advanced camera motions, reflective and refractive materials

Optimising Lattice-Boltzmann for Isambard 3

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- Optimised an implementation of the lattice-boltzmann algorithm to efficiently run on the Isambard 3 Supercomputer
- Implemented serial and vectorised optimisations, OpenMP and MPI, causing speedups of over 500x

Skills & Interests

Programming Languages: Java, C++, C, Python, C#, Javascript, HTML, CSS

Technical Skills: AWS, Git, Unity, Linux

Interests: Monster Hunter, Pokemon, Animated Films, Fudge making, Dinosaurs